

# URBAN NAVIGATION

## Description:

A drone has to navigate to a target and possibly interact with it, but cannot “normally” navigate to the target because of urban obstacles. This could be a macro-scale environment: the drone must navigate through city blocks down alleys and streets while dodging features like lampposts, mailboxes, and cars. It could also be a micro-scale environment: one floor of an office building or similar, where the drone must navigate around walls, pillars, cubicles, and furniture to get to the target.

This scenario is relevant because wars are no longer fought entirely in the open as classical modes of war would dictate. As the battlefield becomes more and more topographically complicated, it becomes increasingly hard to effectively maneuver aerial vehicles, especially UAVs. The existence of physical obstacles also necessitates the need for a method of remotely interacting with objects that are otherwise inaccessible (e.g. someone injured in a remote location that needs medical supplies). UAVs are the perfect solution to problems like these, and it will be useful to develop and test strategies for how to maximize their efficacy in these situations.

## Environment:

- Crowded building interior with target in some arbitrary location, hidden / blocked from normal access by furniture, large objects, walls, etc.
- Crowded city environment with target in arbitrary location, hidden / blocked from normal access by city features, i.e. buildings, lampposts, mailboxes, etc.

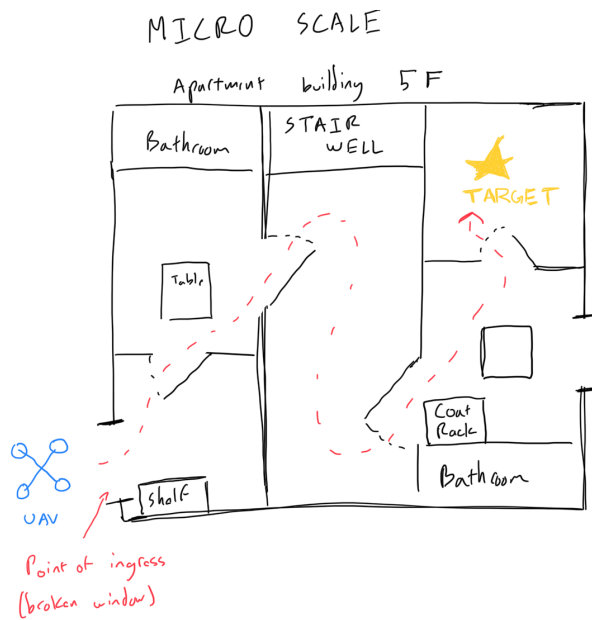
## Entity List:

- Drone (agent)
- Target (providing GPS position, or otherwise recognizable by drone sensor package)
  - An idea that I have had experience with in the past would be an IR beacon that the drone could lock onto and move towards
- Building walls / building interior items (furniture, large objects)
- Building exteriors (forming alleys/urban landmarks)

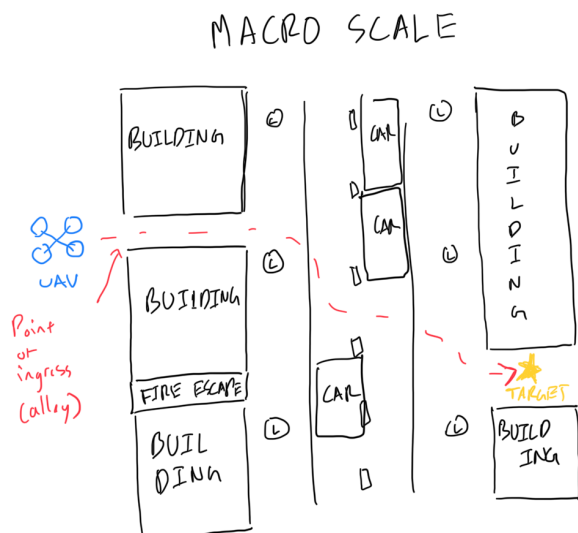
## States:

- Initial state: Drone outside “maze” environment (building interior or collection of exterior buildings)
  - Additional initial state: Drone already inside “maze” environment, but separated from target location
- Goal state: Drone interacting with target at target location (“interact” in the loosest sense of the world”

## Visualization:



- Initial state of micro-scale environment



- Initial state of macro-scale environment

## Actions/Rewards:

- "Finding" a target inside and adding it to the internal map - positive reward
- Navigating to target successfully and performing interaction - positive reward
- Colliding with obstacles/wall - negative reward
- Entering a "stuck" position where no movement is performed for a lengthy duration - negative reward

## Variations:

Some variations of this scenario could take place in similarly crowded environments, such as mountainous regions or forests. The primary idea is that it should be an environment in which straightforward navigation is denied by the presence of multiple obstacles preventing normal flight patterns.

